CT ADEQUACY ANALYSIS REPORT

Device: SIEMENS 7SJ85

Report Ref: CTA-20260729

Date Issued: July 29, 2026

Revision	Prepared By	Checked By	Approved By
Rev. 0	To Be Confirmed	To Be Confirmed	To Be Confirmed

1.0 PROJECT & CLIENT INFORMATION

Client Name: Confidential Client	**Project / Site Company:** Project Site Company Name Not Specified
Project Start Date: To Be Confirmed	**Expected Completion Date:** To Be Confirmed

2.0 EXECUTIVE SUMMARY

This report presents the current transformer (CT) adequacy assessment for "SIEMENS 7SJ85" (Protection Function: SIEMENS 7SJ85). The purpose of this check is to confirm whether the installed/selected CT can deliver sufficient secondary voltage (Vk) / accuracy limit factor to drive connected protection relays correctly under system fault conditions, without core saturation. Based on the inputs and calculations detailed in this report, the CT meets the required knee-point voltage (Vk) / accuracy limit factor and is suitably dimensioned for the fault conditions evaluated. All input data, formulas applied, intermediate values and final results are presented as actually computed for this device, in full, for independent verification by engineering and review teams.

SUITABLY DIMENSIONED

Vk Required: 0 V | Vk Available: N/A

CT Adequacy Analysis Report (continued)

3.0 INPUT PARAMETERS (AS ENTERED)

Parameter	Value	Unit	Description
CT Ratio	**3150/1**	A	Current transformer turns ratio (primary/secondary ampere rating). Determines the scaling factor between primary and secondary circuits.
Accuracy Class	**5P20**		CT accuracy classification per IEC or IEEE standards, indicating the maximum permissible error under specified conditions.
CT Resistance	**9**	ohm	Secondary winding resistance of the current transformer, affecting the voltage drop across the CT itself.
Lead Resistance	**0.54**	ohm	Total loop resistance of the pilot wire/cable connecting the CT secondary to the relay, as computed from the route length and cable data provided.
Relay Burden	**0.04**	VA	Apparent power consumption of the connected relay or meter at rated secondary current, expressed in volt-amperes.
Bus Voltage	**132**	kV	System operating voltage at the point where the CT is installed, used for fault level calculations.
Fault Level	**31.5**	kA	Maximum fault current that can occur at the CT location, determining the worst-case secondary current for adequacy assessment.
Route Length	**1.74**	km	Physical distance from CT to relay location, used to compute lead resistance.

4.0 CALCULATION BREAKDOWN — SIEMENS 7SJ85

Calculation Step / Quantity	Formula Applied (as computed)	Computed Value
1. Max Through Fault Current (Itkmax)	Itkmax = Bus Fault Level (kA) × 1000	**31500 A**
2. CT Primary Current Rating (Ipn)	Ipn = Nameplate Primary Amperes	**3150 A**
3. CT Secondary Resistance (Rct)	Rct = Winding Resistance at 75°C	**9**
4. CT Internal Burden (PE)	PE = $In^2 \cdot Rct$	**9.00 VA**
5. External & Lead Burden (PL)	PL = Lead Burden + Connected Devices	**0.58 VA**
6. CT Rated Burden (PN)	PN = CT Nameplate Burden	**7.5 VA**
7. Accuracy Limit Factor (n)	n = Class ALF factor (e.g. 5P20)	**20**
8. Required Kssc'	**Kssc'(req) = Itkmax / Ipn**	**6.25**
9. Available (Effective) Kssc'	**Kssc'(avail) = n · [(PE + PN) / (PE + PL)]**	**36.64**

FINAL RESULT: Required Kssc' = 6.25 Available Kssc' = 36.64

5.0 NOTES & ASSUMPTIONS

All resistance and current values shown are as computed by the calculation engine from the inputs provided; none are default or placeholder values.

Lead resistance is computed from the route length and cable data supplied for this device/circuit.

Vk Available reflects the manufacturer-declared or test-certificate value supplied for this device; where unavailable, the verdict is reported as NOT APPLICABLE.

This assessment covers CT knee-point voltage / Kssc adequacy only and does not constitute a complete protection coordination study.

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CT Adequacy Assessment — Engineering Report

CT Adequacy Analysis Report (continued)

6.0 FORMULAS & CALCULATION METHOD (KSSC METHOD)

CT adequacy is verified using the Accuracy Limit Factor (Kssc) method per Siemens 7SJ85 requirements:

1. Required Accuracy Limit Factor Kssc':

$Kssc' = Itkmax / Ipn$

Where: Itkmax = Max. through fault current (31500 A), Ipn = CT primary current (3150 A)

2. CT Internal Burden (PE):

$PE = In^2 \cdot Rct$

Where: In = Rated secondary current (1 A), Rct = CT winding resistance (9 ©)

3. Available (Effective) Kssc':

$Kssc'(avail) = n \cdot [(PE + PN) / (PE + PL)]$

Where: n = ALF (20 ratio), PN = Rated burden (7.5 VA), PL = Lead + connected burden (0.58 VA)

4. Verdict Criteria:

' SUITABLE: Available Kssc' "e Required Kssc'
' UNDER-DIMENSIONED: Available Kssc' < Required Kssc'

7.0 APPLICABLE STANDARDS & REFERENCES

IEC 61869-2 Instrument transformers: Additional requirements for current transformers.

IS 2705 (Part 1 to 4) Current transformers: General/specific requirements.

IEEE C37.110 Guide for application of current transformers used for protective relaying purposes.

IEEE C57.13 Standard requirements for instrument transformers.

Internal engineering practice for CT knee-point voltage (Vk) adequacy verification.

8.0 REVIEW & APPROVAL

To Be Confirmed	To Be Confirmed	To Be Confirmed
PREPARED BY	**CHECKED BY**	**APPROVED BY**
Date: _______________	Date: _______________	Date: _______________